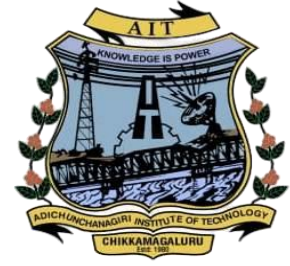


VISVESVARAYA TECHNOLOGICAL UNIVERSITY

“Jnana Sangama”, Belagavi-590 018



A Mini - Project Report

On

**“STUDENT PLACEMENT PREPARATION
PORTAL”**

Submitted in partial fulfillment of the requirements for the **MINI PROJECT (BCD586)**
course of the 5th semester

Bachelor of Engineering

In

Computer Science & Engineering (DATA SCIENCE)

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CERTIFICATE

This is to certify that the Mini project work entitled “**STUDENT PLACEMENT PREPARATION PORTAL**” is a bonafied work carried out by **Ms.Ayesha S (4AI23CD005), Ms.Ayesha Siddiqua (4AI23CD006), Ms.Bhoomika K (4AI23CD008), Ms.Kaivalya Sudhir (4AI23CD021)** in partial fulfillment for the **Mini Project (BCS586)** course of 5th semester Bachelor of Engineering in **Computer Science and Engineering (Data Science)** of the Visvesvaraya Technological University, Belagavi during the academic year **2025-2026**. It is certified that all corrections and suggestions indicated for Internal Assessment have been incorporated in the report deposited in the department library. The Mini project report has been approved as it satisfies the academic requirements in respect of Project Work prescribed for the said Degree.

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ABSTRACT

Student placement preparation plays a vital role in shaping the career opportunities of undergraduate students. However, traditional methods of sharing study materials, collecting alumni feedback, and announcing campus recruitment drives are often unorganized, time-consuming, and inefficient. Students frequently struggle to access relevant resources, company-specific information, and real interview experiences in a structured manner, which affects their placement readiness. The Student Placement Preparation Portal is designed to address these challenges by providing a centralized, user-friendly digital platform that connects students, alumni, and placement coordinators within institution. The project implements a modern and responsive web-based solution using Django (Python) for backend processing, HTML/CSS for UI design, and SQLite for database management. Key functionalities include user authentication, company listings, study material access, alumni feedback posting, and a structured dashboard for each user role (Admin, Student, Alumni).

The project provides an efficient, collaborative, and automated approach to placement preparation. By streamlining information flow and enhancing accessibility of study resources, the portal contributes to better student readiness, improved guidance, and an organized placement preparation ecosystem within educational institutions.

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We express our humble pranamas to his holiness **Divine Soul Parama Poojya Jagadguru Padmabushana Sri Sri Sri Dr. Balagangadharanatha Maha Swamiji** and **Parama Poojya Jagadguru Sri Sri Sri Dr. Nirmalanandanatha Maha Swamiji Pontiff, Sri Adichunchanagiri Maha Samsthana Matt and Sri Sri Gunanatha Swamiji**, Chikkamagaluru branch, Sringeri who have showered their blessings on us.

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CONTENTS

ABSTRACT	i
ACKNOWLEDGEMENTS	ii
CONTENTS	iii
LIST OF FIGURES	iv
LIST OF TABLES	v
LIST OF SNAPSHOTS	vi

CHAPTERS	PAGE NO
1. Introduction	01
1.1 Background	01
1.2 Problem Statement	01
1.3 Objectives of the system	01
1.4 Significance of the system	02
1.5 Scope of the project	02
1.6 Methodology	03
1.7 Target Audience	03
2. System Design	04
2.1 System Architecture	04
2.2 Module Design	04
2.3 Database Design	05
2.4 User Interface (UI) Design	05
2.5 Technology Stack	
3. Implementation	07
3.1 Backend Implementation	07
3.2 Frontend Implementation	08
3.3 Database Implementation	09
4. Testing	11
4.1 Testing Objectives	11

4.2 Testing Environment	11
4.3 Types of Testing	11
4.4 Test Cases	12
5. Results and Discussions	14
5.1 Results	14
5.2 Discussion	19
6. Conclusion and Future Enhancements	21
6.1 Conclusion	21
6.2 Future Enhancements	21
References	22

List of Figures

Sl.No	Description	Page No
2.1	System Architecture	06

List of Tables

Sl.No	Description	Page No
4.1	Test Cases	08

List of Snapshots

Sl. No	Description	Page No
5.1.1	Landing Page	09
5.1.2	Register Page	09
5.1.3	Login Page	10
5.1.4	Company Login Page	10
5.1.5	Study Material Page	11
5.1.6	Alumni Feedback Page	11
5.1.7	Admin Upload Page	12
5.1.8	Admin Feedback Page	12

Chapter 1

INTRODUCTION

1.1 Background

Context:

- Campus placements play a crucial role in shaping the future careers of students. To prepare effectively, students need access to study materials, company-specific interview questions, and guidance from alumni who have already gone through the recruitment process.
- However, this information is often scattered across multiple platforms such as WhatsApp groups, Telegram channels, and email threads.

Problem:

- Manual sharing of files and placement information is unorganized, time-consuming, and often unreliable. Students struggle to find company-wise preparation material when needed. There is no centralized system that stores study notes, alumni feedback, and placement-related updates.

Opportunity:

- A digital portal that centralizes all preparation resources can help students prepare systematically, improve their confidence, and increase their selection chances in campus drives.

1.2 Problem Statement

Overview of the Problem:

- Students lack a unified platform that provides reliable, organized, and company specific placement preparation materials. The manual sharing of files makes it difficult for students to prepare efficiently.

Specific Issues:

- Study materials are scattered across multiple channels.
- No structured system for accessing alumni feedback.
- Students cannot easily find company-wise interview patterns.
- Manual sharing wastes time and leads to loss of important information.

- No proper categorization of aptitude, technical, and HR preparation content.

1.3 Objective of the System

The primary objective of the Student Placement Preparation Portal is to build a **centralized, organized, and easy-to-use platform** for placement preparation.

Key Goals

- Provide structured aptitude, technical, and HR round materials.
- Provide company-specific preparation content (Infosys, TCS, Wipro, etc.).
- Allow alumni to share interview experiences.
- Enable admins to upload materials easily.
- Reduce dependency on external channels like WhatsApp.
- Improve student readiness and confidence for campus drives.

1.4 Significance of the System

- **Centralization:**
All placement-related materials stored in one organized portal.
- **Accuracy:**
Reliable, verified notes and alumni feedback improve preparation quality.
- **Real-Time Access:**
Students can instantly access updated company information and materials.
- **Better Decision Making:**
Students get insights into different company recruitment patterns.
- **Productivity:**
Saves time and reduces confusion caused by multiple sources.

1.5 Scope of the Project

In Scope:

- Student dashboard with categorized study materials.
- Alumni feedback submission and display.
- Company-wise preparation content.

- Admin-side upload of materials and companies.
- Basic authentication (student/admin).

Out of Scope:

- Online coding compiler.
- Automated mock tests (future enhancement).
- Resume analyzer (future enhancement).
- Mobile application version.

1.6 Methodology

Approach:

- The system is developed as a web-based application using HTML, CSS, Django, and SQLite.

Agile Development:

- Iterative design and development with continuous testing and feedback.

Testing:

- Includes unit testing, integration testing, and functional testing to ensure reliability.

1.7 Target Audience

Students:

- Access study materials, company info, and alumni feedback.

Alumni:

- Share company experiences and interview tips.

Administrators:

- Upload materials, manage companies, approve feedback, and maintain portal content.

Chapter 2

SYSTEM DESIGN

The complete design of the Student Placement Preparation Portal, including the system architecture, module structure, database design, user interface screens, and technology stack used. It describes how different components of the system interact with each other and how the system was structured to provide smooth navigation and effective performance.

2.1 System Architecture

High-Level Overview:

- The portal follows a client-server architecture where students access the system through a web interface. The backend (Django) processes requests and retrieves data from the database.

Components:

- **Frontend:** Student dashboard, pages for materials, companies, and feedback.
- **Backend Server:** Handles file uploads, feedback approval, and data retrieval.
- **Database:** Stores company details, study materials, and alumni feedback.

2.2 Module Design

2.2.1 User Authentication Module

- Login for students and administrators.
- Role-based access control.

2.2.2 Study Material Module

- Categorized materials (aptitude, technical, HR).
- Company-wise notes.
- Easy file retrieval.

2.2.3 Alumni Feedback Module

- Alumni can submit interview experiences.
- Admin reviews before publishing.

2.1.4 Company Information Module

- Company details such as rounds, difficulty, and process.

2.3 Database Design

Database tables include:

- **User Table:** user_id, username, password_hash, role
- **Company Table:** company_id, company_name, description
- **Material Table:** material_id, title, category, file_path
- **Feedback Table:** feedback_id, alumni_name, company, feedback_text, status.

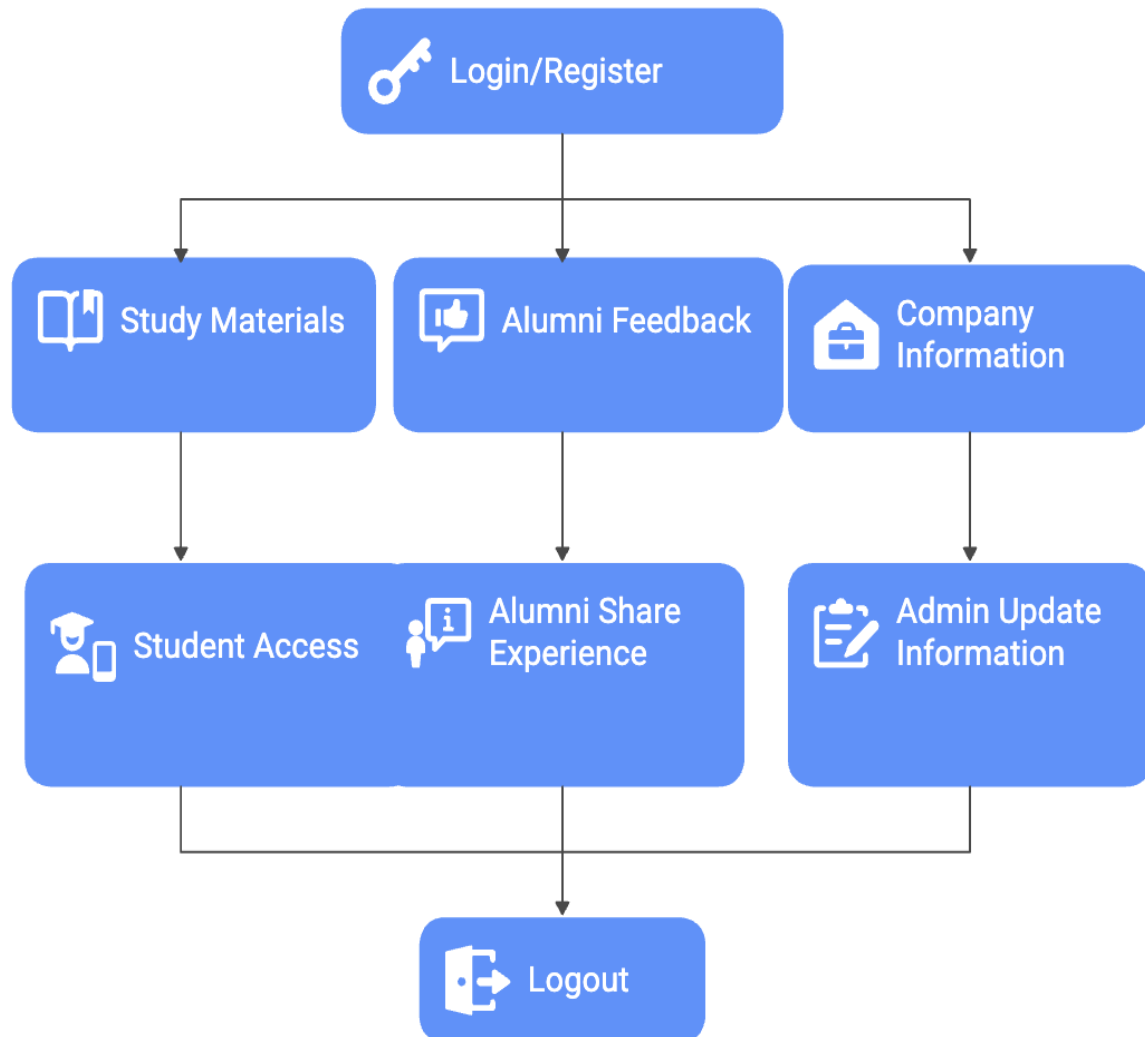
2.4 User Interface (UI) Design

Main Screens:

- Login Screen
- Student Dashboard
- Company List Screen
- Study Materials Screen
- Feedback Screen
- Admin Dashboard

2.5 Technology Stack

- **Frontend:** HTML, CSS
- **Backend:** Django (Python)
- **Database:** SQLite
- **Version Control:** GitHub
- **Server:** Localhost

**Fig 2.1 : System Architecture**

Chapter 3

IMPLEMENTATION

The step-by-step implementation of the portal, covering backend development using Django, frontend development using HTML and CSS, database setup, file upload handling, routing, and user authentication.

3.1 Backend Implementation

The backend is built using the Django framework because it offers a powerful MVC-based structure, built-in security, and an easy-to-use admin interface.

Key Backend Components

Django for Routing, Models, and Admin Panel

Django's URL routing maps every user request to a specific view function.

Examples:

- `/login` → Login View
- `/dashboard` → Dashboard View
- `/materials/<company_id>` → Materials View

This makes navigation consistent and manageable. Django's model layer is used to define the structure of data stored in the database. Each model represents a table with fields such as company name, material file, and feedback text. The Django admin panel helps the administrator manage all backend data without manually writing SQL queries.

Models for Companies, Materials, and Feedback

Three major models were created:

- **Company Model:** Stores company details like name, description, and category.
- **Material Model:** Stores study materials uploaded by the admin. Includes fields such as material name, file path, description, and company reference.
- **Feedback Model:** Stores alumni experiences, interview stories, and preparation tips. Includes fields like feedback text, user name, and status (approved/pending).

Admin Functionalities

The admin user has access to additional features:

- Add or edit company details
- Upload new study materials in PDF, PPT, or video formats
- Approve or reject feedback submitted by users
- View the number of downloaded materials and feedback summaries

Backend APIs for Fetching Company-wise Materials

Custom API endpoints are created to return data to the frontend.

These APIs provide:

- List of companies
- Study materials filtered by company
- Approved feedback
- User-specific data if needed

3.2 Frontend Implementation

The frontend is created using HTML, CSS, and Django templates. It focuses on readability, easy navigation, and a clean layout so that students can quickly find relevant study materials.

UI Components

Login Page

- Accepts username and password
- Validates inputs using Django's authentication system
- Redirects users to the dashboard on successful login
- Displays error messages for incorrect credentials

Dashboard

The dashboard acts as the main entry page after login.

It displays:

- Categories such as Aptitude, Coding, Interview Tips
- List of companies for which materials are available
- Navigation links to materials, feedback, and logout

Study Material Page

This page shows all study materials linked to a selected company.

It includes:

- Material titles
- Descriptions
- Download buttons
- File formats (PDF/PPT/Video)

Materials are grouped under different categories to help students quickly access what they need.

Feedback Page

Displays alumni experiences and interview journeys. The feedback is shown only after admin approval.

Users can:

- Read interview tips
- Browse stories company-wise
- Gain insights about hiring rounds

Admin Pages

These pages support portal maintenance:

- Add new companies
- Upload materials with file attachments
- Approve or delete feedback
- View list of all uploaded files.

Admin pages are protected using Django's permission controls to prevent unauthorized access.

3.3 Database Implementation

The database acts as the backbone of the system, storing all essential information such as companies, study materials, user data, and feedback entries.

SQLite Used as Backend Database

SQLite is the default database for Django and is suitable for development-based projects. It is lightweight, fast, and does not require a separate server setup.

Relational Design Ensures Data Consistency

The database uses a relational structure where tables are linked through primary keys and foreign keys. This ensures:

- Non-duplicate company entries
- Proper linking between materials and their companies
- Accurate retrieval of approved feedback

Foreign Keys for Linking Materials and Feedback

Foreign keys are used for:

- **Materials→Companies**
Each study material belongs to a specific company.
- **Feedback→Users/Companies**
Feedback is mapped to either a company or a user.

Chapter 4

TESTING

The various testing methods used to ensure that the Student Placement Preparation Portal functions correctly and reliably. It describes the testing environment, test objectives, types of tests performed (unit, integration, functional), and the tools used for testing.

4.1 Testing Objectives

- Validate that students can access study materials correctly.
- Ensure admin uploads appear in the right categories.
- Verify feedback submission and approval flow.
- Check UI responsiveness and navigation.

4.2 Testing Environment

Hardware: Laptop/PC with 8GB RAM

Software:

- Backend: Django (Python)
- Frontend: HTML/CSS
- Database: SQLite
- Tools: Chrome browser

4.3 Types of Testing

4.3.1 Unit Testing

- Login functionality
- Material upload function
- Feedback save operation

4.3.2 Integration Testing

- Connection between frontend and backend
- File upload linkage with database.

4.3.3 Functional Testing

- Company-wise material display
- Feedback approval visibility
- Navigation and UI flow

4.4 Test Cases

Below are test cases for various components:

Table 4.1 Test Cases

Test Case ID	Description	Test Steps	Expected Result	Status
TC-01	Login with valid credentials	1. Enter valid username & password 2. Click “Login”	User is redirected to their dashboard	Pass
TC-02	Login with invalid credentials	1. Enter wrong username or password 2. Click “Login”	Error message: “Invalid credentials”	Pass
TC-03	Student views study materials	1. Login as Student 2. Navigate to “Study Materials”	Materials displayed correctly	Pass
TC-04	Student views company information	1. Login as Student 2. Click “Companies” 3. Select a company	Company details displayed	Pass
TC-05	Student views alumni feedback	1. Login as Student 2. Open “Feedback” page	Approved Feed back entries displayed	Pass
TC-06	Unauthorized access attempt	1. Login as Student 2. Try accessing admin URL manually	System shows “Access denied”	Pass
TC-07	Students filters materials by category	1. Open Materials page 2. Select category (Aptitude/Technical/HR)	Only filtered materials displayed	Pass

TC-08	Admin adds new company information	1. Login as Admin 2. Open “Add Company” page 3. Fill form → Submit	Company appears in company list	Pass
TC-09	Alumni submits interview feedback	1. Open Feedback page 2. Enter details 3. Click Submit	Feedback saved with Pending status	Pass
TC-10	Admin approves alumni feedback	1. Login as Admin 2. Open “Feedback Approvals” 3. Approve a feedback entry	Feedback becomes visible to students	Pass
TC-11	Student searches	1. Login 2. Enter keyword in search bar 3. Click “Search”	Search results displayed based on keyword	Pass

Chapter 5

RESULTS AND DISCUSSION

The entire project, restating how the portal meets its objectives and supports students in preparing for placements. It highlights the advantages of having a centralized platform and its impact on improving preparation efficiency and student confidence.

5.1 Results

Snapshots of the Project with description:

- **Landing Page:** This is the home or landing page of the portal. It provides an entry point for users where they can choose to log in or explore the portal.

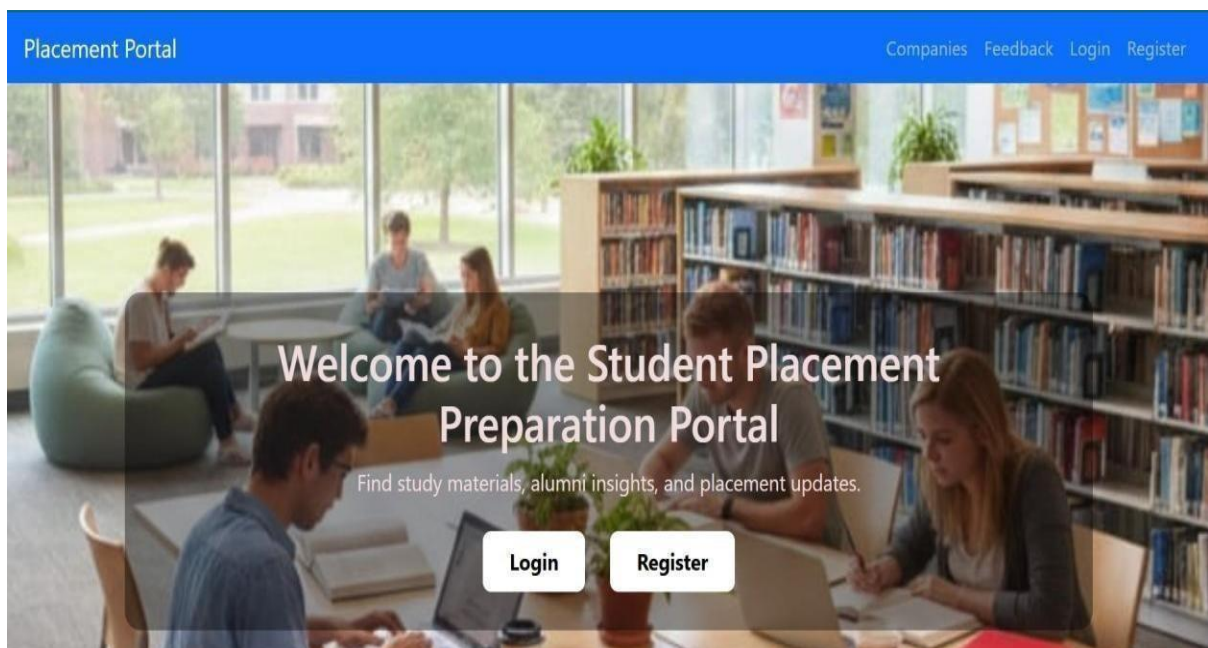
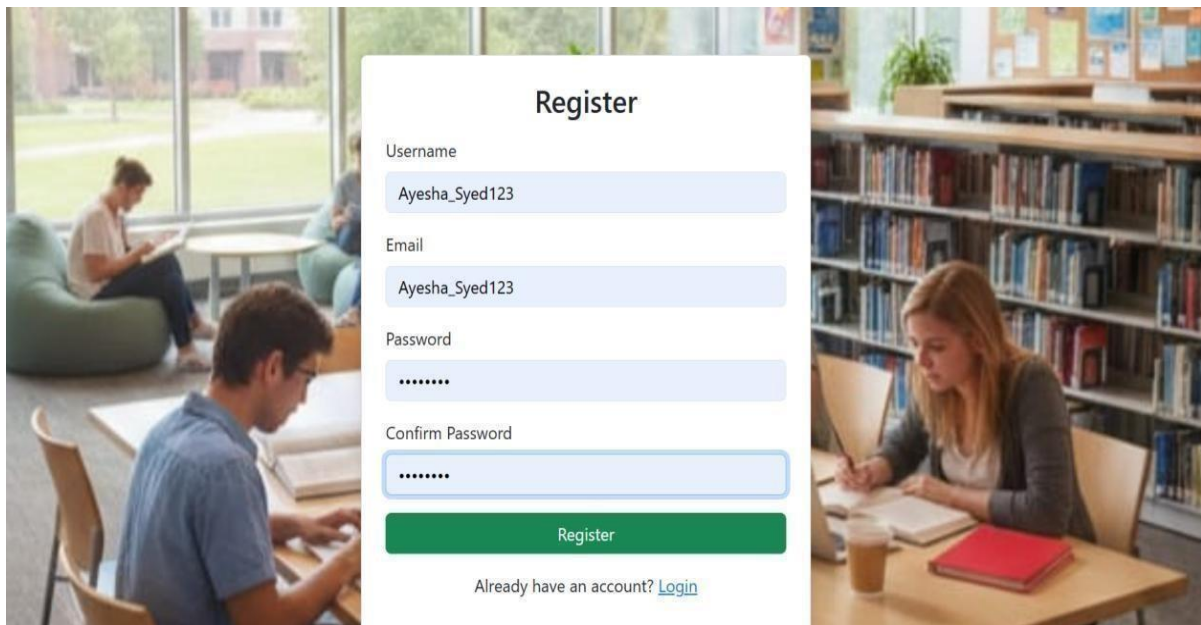


Fig 5.1.1 Landing Page

- **Register Page :** The registration page allows new students or alumni to create an account. Users enter their details, which are saved in the database. This ensures only authorized members can access study materials and submit feedback.

The image shows a 'Register' form overlaid on a background of students in a library. The form has a title 'Register' at the top. It contains four input fields: 'Username' with the value 'Ayesha_Syed123', 'Email' with the value 'Ayesha_Syed123', 'Password' with masked characters '.....', and 'Confirm Password' with masked characters '.....'. Below the fields is a green 'Register' button. At the bottom, there is a link that says 'Already have an account? [Login](#)'.

Register

Username
Ayesha_Syed123

Email
Ayesha_Syed123

Password
.....

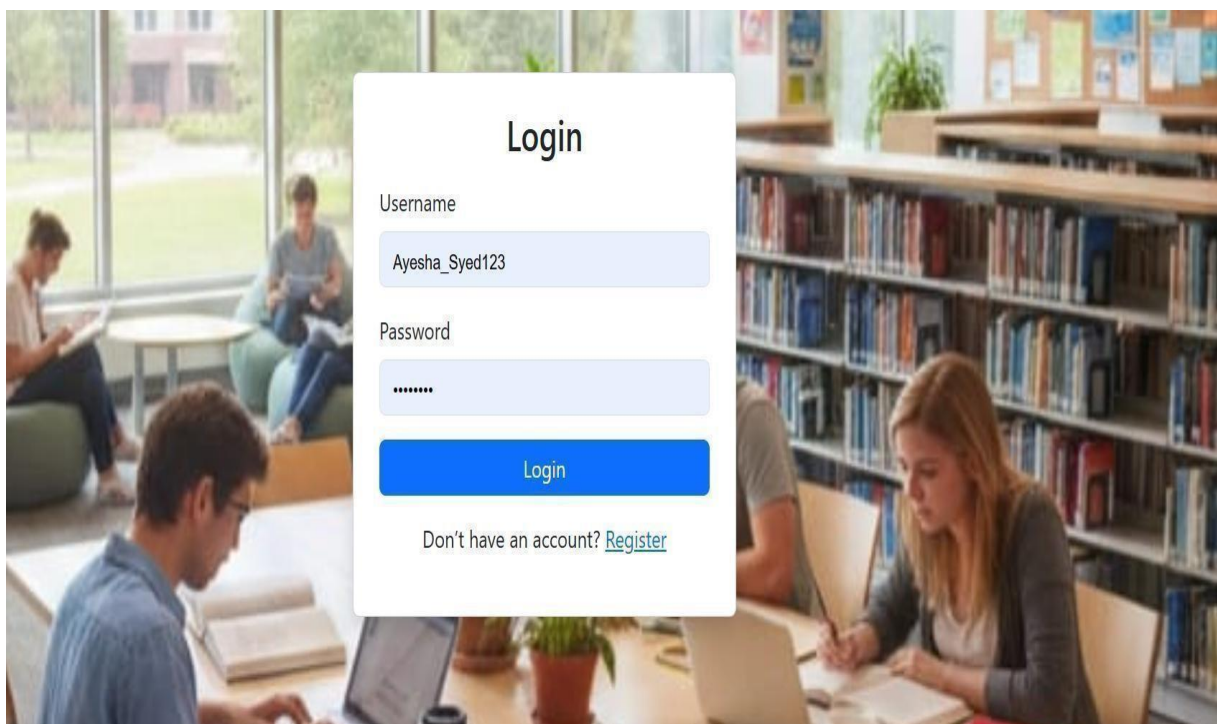
Confirm Password
.....

[Register](#)

Already have an account? [Login](#)

Fig 5.1.2 Register Page

- **Login Page:** The login page provides secure access to the portal. Users must enter valid credentials

The image shows a 'Login' form overlaid on a background of students in a library. The form has a title 'Login' at the top. It contains two input fields: 'Username' with the value 'Ayesha_Syed123' and 'Password' with masked characters '.....'. Below the fields is a blue 'Login' button. At the bottom, there is a link that says 'Don't have an account? [Register](#)'.

Login

Username
Ayesha_Syed123

Password
.....

[Login](#)

Don't have an account? [Register](#)

Fig 5.1.3 Login Page

- **Company List Page:** This page displays the list of companies available in the portal. Students can select a company to view its recruitment process, difficulty levels, commonly asked questions, and alumni experiences.

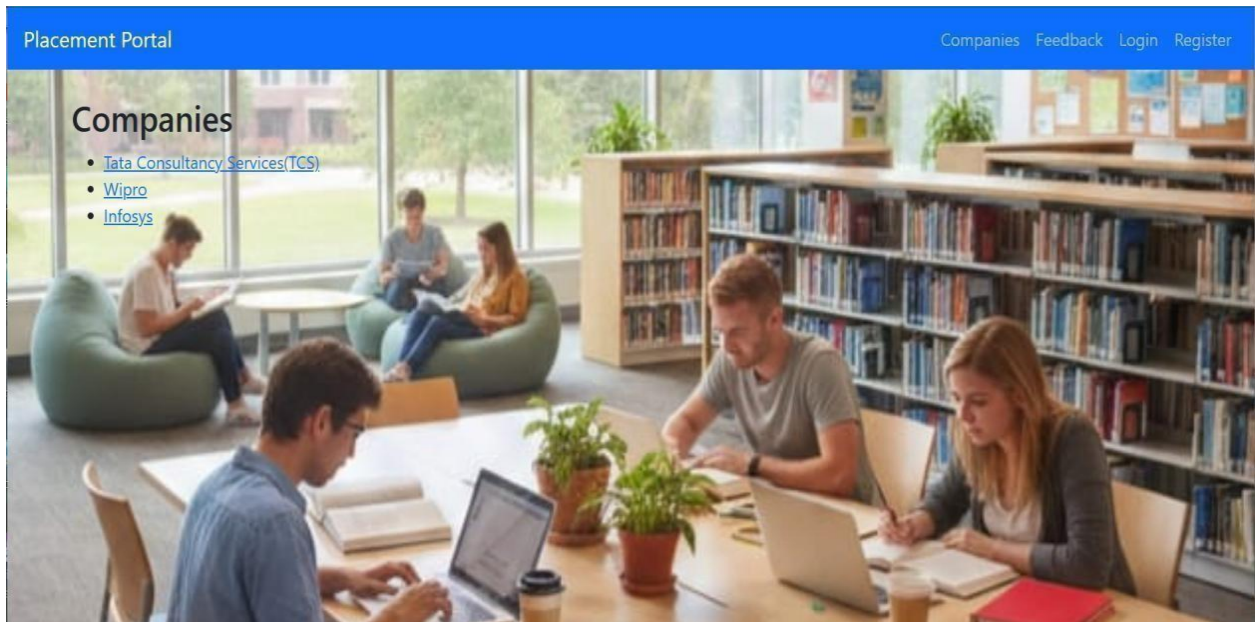


Fig 5.1.4 Company List Page

- **Study Materials Page:** This page shows categorized placement preparation materials. Students can access Aptitude, Technical, HR, or company-specific documents. Materials are uploaded by the admin and organized for easy navigation.

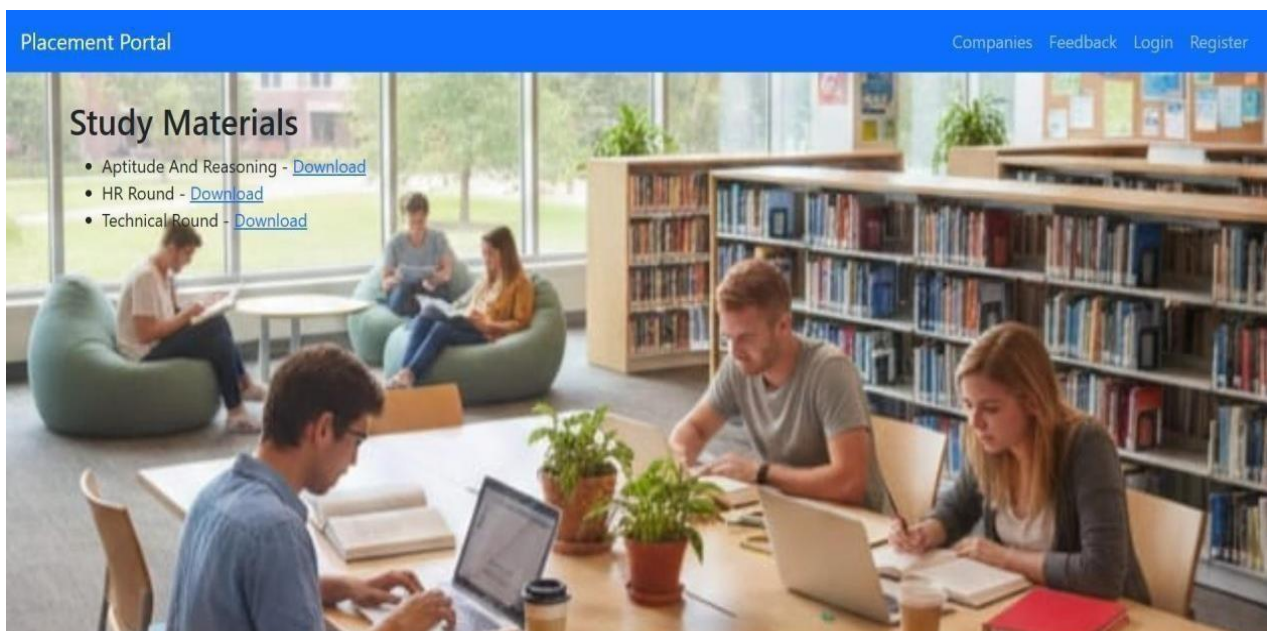


Fig 5.1.5 Study Materials Page

- **Alumni Feedback Page :** This page lists approved alumni feedback entries. Students can read real experiences shared by seniors, helping them understand actual interview scenarios and improve their preparation strategy.

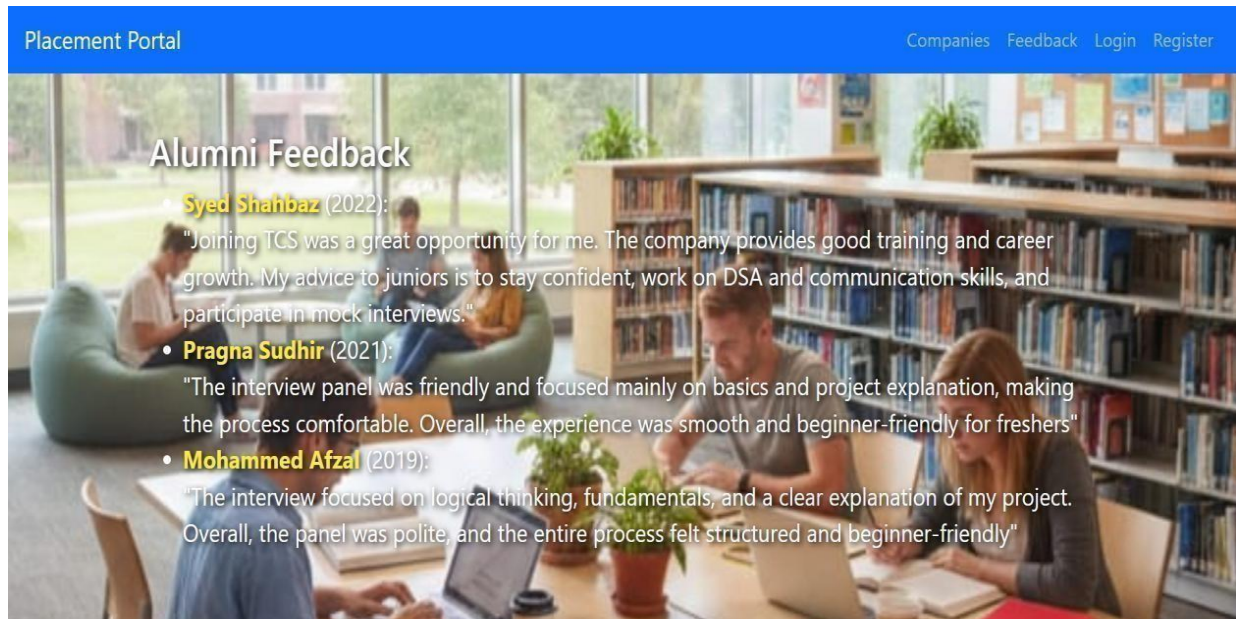


Fig 5.1.6 Alumni Feedback Page

- **Admin Upload Page:** This section allows admins to upload new study materials, documents, or resources.

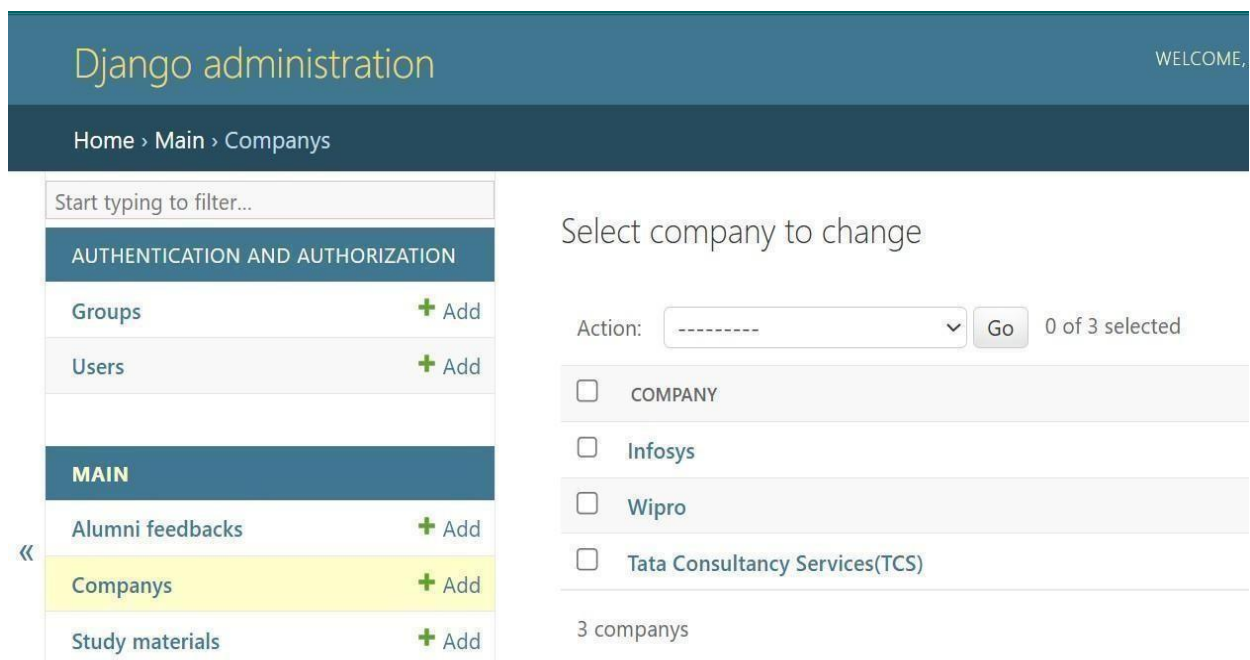


Fig 5.1.7 Admin Upload Page

- **Admin Feedback Page:** This page is used by admins to approve or reject alumni feedback. Only approved feedback is visible to students, ensuring authenticity and reliability of shared experiences.

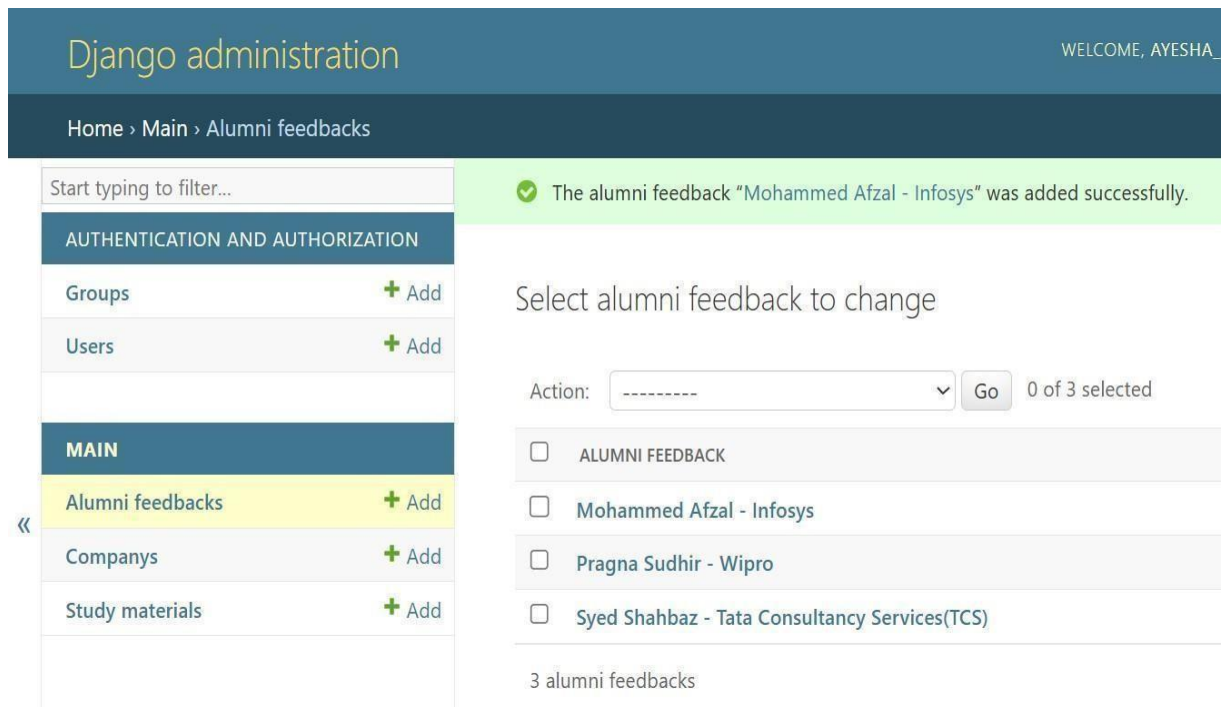


Fig 5.1.8 Admin Feedback Page

5.2 Discussion

Effectiveness of the System

The portal effectively meets the goals outlined in the problem statement and objectives:

Centralized Access to Resources

- One of the primary achievements of the system is the integration of all placement related materials—aptitude, technical, HR, and company-wise resources—into a single platform. This eliminates the need for students to rely on unorganized WhatsApp groups, drives, or email chains.

Improved Student Preparation by providing:

- Structured study materials,
- Company-specific information,
- Alumni experiences.

Challenges Encountered

During development, several challenges arose:

Organizing a Large Volume of Study Materials

Study materials had to be separated into:

- Aptitude
- Technical
- HR
- Company-wise

Ensuring Authenticity of Alumni Feedback

- Alumni submissions had to be reviewed before publishing to maintain the credibility of the content. This required creating an approval workflow in the admin module.

Maintaining a Clean and Non-Overwhelming UI

- Because the portal contains multiple sections, it was challenging to design a layout that remained simple while still presenting all necessary information.

Managing File Upload Formats and Sizes

- Implementing validation to avoid invalid or oversized files was necessary for smooth functioning.

Limitations of the Current System

Despite the system's successful performance, some limitations were identified:

- No integrated mock test or quiz module
- No mobile application available
- No AI-based recommendation or resume analyzer
- Limited automation in sending updates or notifications
- Manual alumni feedback entry instead of automated verification

Solutions Implemented

Improved Categorization Structure

- Study materials were grouped efficiently with clear navigation menus and filters, making it easy for students to access any content.

Admin Approval for Feedback

- A review mechanism was added so that only approved feedback becomes visible to students, ensuring reliability and quality.

File Validation for Secure Uploads

- Only specific file types and sizes are allowed during material upload, preventing errors and ensuring security.

Chapter 6

CONCLUSION AND FUTURE ENHANCEMENT

6.1 Conclusion

The **Student Placement Preparation Portal** effectively addresses the major challenges faced by students during placement preparation, including scattered study materials, lack of structured guidance, unverified alumni feedback, and difficulty in accessing company-specific resources. By replacing unorganized manual sharing methods with a centralized digital system, the project enhances accessibility, saves preparation time, and improves the overall learning and placement readiness of students.

Key achievements of the system include:

- **Centralized Learning Resources:** All aptitude, technical, and HR materials are organized in one platform, making preparation easier and more structured.
- **Company-Wise Preparation Support:** Students receive detailed insights into interview patterns, selection rounds, and expectations of different companies.
- **Verified Alumni Feedback:** Authentic experiences from alumni help students understand real interview environments and boost confidence.
- **Simplified Material Management:** Admins can easily upload study content, company information, and approve feedback, ensuring the portal stays updated.
- **User-Friendly Interface:** The clean and responsive design ensures smooth navigation for students, alumni, and administrators.

6.2 Future Enhancements

- Add mock tests and coding challenges.
- Provide resume builder with ATS score.
- Introduce mobile application support.
- Add gamification features like badges and leaderboards.
- Implement real-time notifications for placement drives.
- Mobile Application for Android & iOS.
- Gamification Features.
- Integration With College Placement Cell.

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